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GRADE GUIDANCE SERIES

GRADE guidance 39: using GRADE-ADOLOPMENT to adopt, adapt or create contextualized recommendations from source guidelines and evidence syntheses

Miloslav Klugar[a,b,1], Tamara Lotfi[c,d,1], Andrea J. Darzi[d,e], Marge Reinap[f], Jitka Klugarová[a,b], Lucia Kantorová[a,g], Jun Xia[h,i], Romina Brignardello-Petersen[d], Andrea Pokorná[a,j], Glen Hazlewood[k], Zachary Munn[l], Rebecca L. Morgan[d,m], Ingrid Toews[n], Ignacio Neumann[d,o], Patraporn Bhatarasakoon[p], Airton Tetelbom Stein[q], Michael McCaul[r], Alexander G. Mathioudakis[s,t], Kristen E. D'Anci[u], Grigorios I. Leontiadis[v], Celeste Naude[r], Lenny T. Vasanthan[w], Joanne Khabsa[x], Malgorzata M. Bala[y,z], Reem Mustafa[d,aa], Karen DiValerio Gibbs[ab], Robby Nieuwlaat[c,d], Nancy Santesso[c,d], Dawid Pieper[ac,ad], Saphia Mokrane[ae,af], Israa Soghier[ag], Wanchai Lertwatthanawilat[p], Wojtek Wiercioch[c,d], Shahnaz Sultan[ah], Jana Rozmarinová[a], Pavla Drapačová[a], Yang Song[ai], Marwa Amer[aj,ak], Yasser S. Amer[al,am], Shahab Sayfi[c,d], Ilse M. Verstijnen[an], Ein-Soon Shin[ao], Zuleika Saz-Parkinson[ap], Kevin Pottie[aq], Alessandra Ruspi[ar,as], Ana Marušić[at], K.M. Saif-Ur-Rahman[au,av], Maria X. Rojas[aw], Elie A. Akl[d,x], Holger J. Schünemann[d,ar,ax,ay,*], for the GRADE Working Group

[a]*Cochrane Czech Republic, Czech CEBHC: JBI Centre of Excellence, Czech GRADE Network, Institute of Health Information and Statistics of the Czech Republic, Prague, Czech Republic*

[b]*Center of Evidence-based Education and Arts Therapies: A JBI Affiliated Group, Faculty of Education, Palacký University Olomouc, Prague, Czech Republic*

[c]*World Health Organization Collaborating Center for Infectious Diseases, Research Methods and Recommendations, Michael G. DeGroote Cochrane Canada & McMaster GRADE Centres, McMaster University, Hamilton, Ontario, Canada*

[d]*Department of Health Research Methods, Evidence and Impact, McMaster University, 1280 Main Street West, Hamilton, L8S 4L8, Ontario, Canada*

[e]*Department of Anesthesia, Faculty of Health Sciences, McMaster University, Hamilton, Ontario, Canada*

[f]*Division of Country Health Policies and Systems, World Health Organization Regional Office for Europe, Copenhagen, Denmark*

[g]*Institute of Biostatistics and Analyses, Faculty of Medicine, Masaryk University, Brno, Czech Republic*

[h]*Agency for Clinical Practice Guideline, Korean Academy of Medical Sciences, Seoul, South Korea*

[i]*Division of Lifespan and Population Health, The University of Nottingham, Nottingham, UK*

[j]*Department of Health Sciences, Faculty of Medicine, Masaryk University, Brno, Czech Republic*

[k]*Department of Medicine, Cumming School of Medicine, University of Calgary, Calgary, Alberta, Canada*

[l]*Health Evidence Synthesis, Recommendations and Impact, School of Public Health, The University of Adelaide, Adelaide, South Australia, Australia*

[m]*School of Medicine, Case Western Reserve University, Cleveland, OH USA*

[n]*Institute for Evidence in Medicine, Medical Center - University of Freiburg, Faculty of Medicine, University of Freiburg, Freiburg, Germany*

[o]*School of Medicine, Universidad San Sebastian, Santiago, Chile*

[p]*Thailand Center for Evidence-Based Health Care, Faculty of Nursing, Chiang Mai University, Chiang Mai, Thailand*

[q]*Department of Public Health, Universidade Federal de Ciências da Saúde de Porto Alegre (Ufcspa), Porto Alegre, Brazil*

[r]*Centre for Evidence-based Health Care, Division of Epidemiology and Biostatistics, Department of Global Health, Stellenbosch Univeristy, Cape Town, South Africa*

[s]*Division of Immunology, Immunity to Infection and Respiratory Medicine, School of Biological Sciences, The University of Manchester, Manchester, UK*

[t]*North West Lung Centre, Wythenshawe Hospital, Manchester University NHS Foundation Trust, Manchester Academic Health Science Centre, Manchester, UK*

[1] Co-first authors.

* Corresponding author. Clinical Epidemiology and Research Center (CERC), Humanitas University & Humanitas Research Hospital, Via Rita Levi Montalcini 4 - 20090 Pieve Emanuele, Milano, Italy.

E-mail address: schuneh@mcmaster.ca (H.J. Schünemann).

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[u]Sigma Health Consulting, LLC, McLean, VA, USA
[v]Department of Medicine, McMaster University, 1280 Main Street West, Hamilton, L8S 4L8, Ontario, Canada
[w]Department of Physical Medicine and Rehabilitation, Christian Medical College, Vellore, India
[x]Department of Internal Medicine, American University of Beirut Medical Center, Beirut, Lebanon
[y]Chair of Epidemiology and Preventive Medicine, Jagiellonian University Medical College, Krakow, Poland
[z]Krakow GRADE Centre, Krakow, Poland
[aa]Division of Nephrology and Hypertension, University of Kansas Medical Center, Kansas City, MO, USA
[ab]Department of Pediatrics, Baylor College of Medicine, Houston, TX, USA
[ac]Faculty of Health Sciences Brandenburg, Brandenburg Medical School (Theodor Fontane), Institute for Health Services and Health System Research, Rüdersdorf, Germany
[ad]Center for Health Services Research, Brandenburg Medical School (Theodor Fontane), Rüdersdorf, Germany
[ae]Department of Primary Care, Faculty of Medicine, Université Libre de Bruxelles, Brussels, Belgium
[af]Working group Development of Primary Care Guidelines, Antwerpen, Belgium
[ag]Department of Medicine, Massachusetts General Brigham/Salem Hospital, Boston, MA, USA
[ah]Division of Gastroenterology, Hepatology and Nutrition, University of Minnesota, Minneapolis, MN, USA
[ai]Iberoamerican Cochrane Centre, Biomedical Research, Institute Sant Pau (IIB Sant Pau), Barcelona, Spain
[aj]Medical/Critical Pharmacy Division, King Faisal Specialist Hospital and Research Center, Riyadh, Saudi Arabia
[ak]College of Medicine and Pharmacy, Alfaisal University, Riyadh, Saudi Arabia
[al]Pediatrics Department, Quality Management Department, King Saud University Medical City, Research Chair for Evidence-Based Healthcare and Knowledge Translation, King Saud University, Riyadh, Saudi Arabia
[am]Department of Internal Medicine, Ribeirão Preto Medical School, University of São Paulo, Ribeirão Preto, São Paulo, Brazil
[an]Research and Development, National Health Care Institute, Zorginstituut, The Netherlands
[ao]Department of Preventive Medicine and Public Health, School of Medicine, Ajou University, Suwon, South Korea
[ap]European Research Executive Agency, European Commission, Brussels, Belgium
[aq]Departments of Family Medicine and Epidemiology and Biostatistics, Western University, London, Ontario, Canada
[ar]Clinical Epidemiology and Research Center (CERC), Humanitas University & Humanitas Research Hospital, Via Rita Levi Montalcini 4 - 20090 Pieve Emanuele, Milano, Italy
[as]Physiotherapy Unit, Humanitas Clinical and Research Center − IRCCS, Rozzano, Milan, Italy
[at]Center for Evidence-based Medicine, Medicinski fakultet Sveucilista u Splitu, University of Split School of Medicine, Split, Croatia
[au]College of Medicine, Nursing, and Health Sciences, University Galway, Galway, Ireland
[av]Evidence Synthesis Ireland and Cochrane Ireland University of Galway, Galway, Ireland
[aw]Department of Clinical Epidemiology and Public Health, The Sant Pau Biomedical Research Institute (IIB-Sant Pau), Barcelona, Spain
[ax]Nottingham Ningbo GRADE Centre, School of Economics, The University of Nottingham Ningbo China, Ningbo, China
[ay]China European Research Executive Agency, European Commission, Brussels, Belgium

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Abstract

Background and Objective: The Grading of Recommendations, Assessment, Development and Evaluations (GRADE)-ADOLOPMENT methodology has been widely used to adopt, adapt, or de novo develop recommendations from existing or new guideline and evidence synthesis efforts. The objective of this guidance is to refine the operationalization for applying GRADE-ADOLOPMENT.

Methods: Through iterative discussions, online meetings, and email communications, the GRADE-ADOLOPMENT project group drafted the updated guidance. We then conducted a review of handbooks of guideline-producing organizations, and a scoping review of published and planned adolopment guideline projects. The lead authors refined the existing approach based on the scoping review findings and feedback from members of the GRADE working group. We presented the revised approach to the group in November 2022 (approximately 115 people), in May 2023 (approximately 100 people), and twice in September 2023 (approximately 60 and 90 people) for approval.

Results: This GRADE guidance shows how to effectively and efficiently contextualize recommendations using the GRADE-ADOLOPMENT approach by doing the following: (1) showcasing alternative pathways for starting an adolopment effort; (2) elaborating on the different essential steps of this approach, such as building on existing evidence-to-decision (EtDs), when available or developing new EtDs, if necessary; and (3) providing examples from adolopment case studies to facilitate the application of the approach. We demonstrate how to use contextual evidence to make judgments about EtD criteria, and highlight the importance of making the resulting EtDs available to facilitate adolopment efforts by others.

Conclusion: This updated GRADE guidance further operationalizes the application of GRADE-ADOLOPMENT based on over 6 years of experience. It serves to support uptake and application by end users interested in contextualizing recommendations to a local setting or specific reality in a short period of time or with limited resources. © 2024 Published by Elsevier Inc.

Keywords: GRADE; Guidelines; Adaptation; Adoption; Adolopment; Systematic reviews; Contextualization; Global health; Research waste

What is new?

Key findings

- The Evidence to Decision (EtD) frameworks and their corresponding criteria are used by many organizations globally.
- The GRADE-ADOLOPMENT methodology, fundamentally based on decision criteria, has proven to be useful, feasible and helpful for guideline developers who want to ensure that recommendations correspond to their own setting and reality.

What this adds to what is known?

- This article provides new GRADE guidance and practical examples for the use of GRADE-ADOLOPMENT, an evidence to decision (EtD) framework based approach to adapting and contextualizing guidelines.

What is the implication and what should change now?

- This guidance refines the previously published GRADE-ADOLOPMENT approach based on a review, experiences and practical lessons learnt.
- On this basis, we provide an updated operationalization of the approach to facilitate replication and uptake by other guideline developers. We also emphasize the need to disseminate and make available adolopment efforts including adapted or de novo EtDs to support use by others.

1. Background

Trustworthy guidelines are the cornerstone for providing evidence-informed recommendations in response to clinical, public health, and health policy questions [1]. Trustworthy guidelines may also be used to support shared decision-making between health care providers and patients to individualize care, and their recommendations can be contextualized to facilitate implementation. Language translation (if applicable) of an existing guideline is rarely sufficient to address context and successfully implement recommendations [2]. Contextualization of an already published guideline's recommendations to a different setting can save both direct and indirect financial or nonfinancial resources compared to *creating new guidelines* (Box 1) [9,10]. The term "contextualization" provides an umbrella term for adaptation in the literature [8,11−13], but in the context of guidelines, it is understood as meaning "that relevant, specific information is considered in each recommendation for the intended setting and context (Box 1)". Thus, contextualization includes what is more narrowly described as adaptation of guidelines that focuses on the use of an existing source guideline and adapting it for a setting without necessarily, including prioritization processes [8,14]. In 2006, we created a simple checklist for the World Health Organization (WHO) to understand the need for adaptation of guidelines, but much work has been done since. Today, adaptation is of utmost importance for guideline developers to support sharing information with relevant decision-makers and streamline development processes to avoid duplication of efforts and resource waste [8]. Box 1 provides an overview of the terms we use in this document and related concepts. A critical clarification that has evolved over time is the distinction between the concept of adaptation of guidelines and single recommendations (Box 1).

The WHO, the Institute of Medicine (now the National Academy of Medicine), the Guidelines International Network (GIN), and the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) working group published sets of standards for trustworthy clinical practice guidelines and the Appraisal of Guidelines for Research and Evaluation (AGREE) collaboration created a guideline quality assessment tool, providing the background for criteria by which a guideline can be judged [3,15−22]. In accordance with these standards, GRADE-ADOLOPMENT now describes an efficient approach for guideline contextualization that allows for local, regional, or national input, as well as stakeholder involvement and ownership while maintaining transparency, rigor, and quality (Box 1) [7]. This involvement and ownership is critical for the acceptability and uptake of recommendations [7,11,12,23−25]. Several frameworks for guideline adaptation and a checklist to identify the need for adaptation exist, but a recent systematic review judged GRADE-ADOLOPMENT to be the most complete and recent [13,26]. To respond to the evolving landscape of guideline contextualization and in comparison with other approaches that focus on existing guidelines to formulate guideline questions, adolopment begins with prioritizing guideline questions to ensure relevance to end users before identifying potential credible source guidelines to support the process. Adolopers may then update the underlying evidence synthesis for the guideline criteria used to determine the strength and direction of recommendations, and use the GRADE evidence-to-decision (EtD) framework, upon which GRADE-ADOLOPMENT is based, to formulate context-specific recommendations with their guideline panel. Adolopment is a neologism referring to the systematic and transparent approach to adoption, adaptation, and/or de novo development of recommendations to fit the context of interest of a guideline using the GRADE methodology [7].

A recent scoping review captured 21 examples of the approach from Europe (6 guidelines), Australia (three), Canada (two), Latin America (three), Mexico (one), Qatar (one),

Box 1 Definitions related to Grading of Recommendations, Assessment, Development and Evaluations(GRADE)-ADOLOPMENT

GRADE

The GRADE approach provides a transparent framework for assessing the certainty of the evidence and developing recommendations according to transparently described criteria [3−6]. GRADE, developed with input from groups such as the WHO, can be used in systematic reviews, health technology assessment, and guidelines (www.gradeworkinggroup.org) and has been adopted by over 110 organizations globally. The GRADE approach characterizes recommendations through their certainty in the supporting evidence, strength (strong or conditional, also called weak), and direction (for or against the intervention).

Source guideline

A source guideline is the guideline from another entity (eg, public health or professional society guideline) that is being considered for contextualization. Typically, a guideline includes a set of actionable statements, the recommendations. Some or all the recommendations of a guideline can be used for contextualization, depending on priorities and question matching done by the group interested in adoloping recommendations.

EtD framework

EtD tables and frameworks provide information on flexible criteria that influence guideline recommendations (eg, problem, health benefits and harms, their balance, certainty in the best available evidence, cost and cost-effectiveness, feasibility, acceptability, equity, and values and preferences) and how the guideline development group or panel can judge the effect of these criteria on the final recommendations [7]. The EtD framework transparently facilitates adolopment of guideline recommendations to the setting, context, health system, and culture of a specific jurisdiction or country.

Contextualization

Contextualization of guidelines ensures that relevant, context-specific information is considered in each recommendation.

Contextualization describes the process of [8]:

- Acknowledging the need for dialogue and formal consideration of context-specific best available evidence and criteria for adopting, adapting, or de novo creation of recommendations from an existing trustworthy source guideline to the national, regional, or local level;
- Deciding whether the recommended interventions are feasible and acceptable for that setting or context and;
- Modifying recommendations or adding to the remarks to optimize their implementation using structured and transparent processes

Adolopment

Adolopment is a process for guideline development that combines adoption, adaptation, and de novo development of recommendations [7]. As an initial step, the adolopment approach begins with the selection of guideline topics, scoping reviews of the available literature, and question prioritization. Relevant, credible and up-to-date recommendations are identified, and their EtDs or information around the EtD criteria (ie, benefits and harms, values and preferences, cost and cost effectiveness, equity, feasibility, and acceptability) are reviewed (or developed if missing in a source guideline) and contextual data added when necessary. A guideline panel may then decide to maintain the source recommendation as is (adopt) or to modify (adapt) the source recommendations. If relevant, credible and up-to-date recommendations are not available for certain prioritized guideline questions, the guideline group may decide to develop a de novo recommendation. A final adoloped guideline may be composed of a mix of adopted, adapted, and de novo recommendations. We will use the term adolopment in the specific context of the GRADE-ADOLOPMENT methodology that refers to the process of guideline contextualization across multiple recommendations to create relevant guidelines. Adolopment can be applied to any type of guideline question (eg, diagnosis, treatment, testing, or management).

Adoption

Adoption means the use of an existing recommendation, either unmodified or with minimal changes. Ideally, this includes that the source guidelines are reviewed and the judgments and decisions on the criteria that led to a specific

recommendation are still agreed upon. For adoption, the source guideline must provide a clear, documented path to the recommendations from the evaluation of the evidence to the preparation of the recommendation itself. If the path is credible and transparent, the entity responsible for implementation must assess the directness and timeliness of the recommendations and the judgments made. Directness refers to the concept that the recommendations are appropriate and applicable to the context of the health-care setting of interest, for example, by addressing populations, interventions, and outcomes of interest. To assess if the information is up to date, it is usually necessary to re-examine the incorporated evidence to ensure that no new or other essential evidence should be added, for example, by updating the corresponding evidence synthesis performed in the original resource.

Adaptation

Adaptation of a recommendation means that reliable recommendations exist that meet the established criteria for credibility but that the judgements on the criteria that support the recommendation, or the recommendation itself, require updates or changes to be implemented for the health-care setting of interest. For example, the new recommendation may need to broaden the population of interest or may need to consider local costs that differ from the source guideline recommendation. Any change should be documented and the rationale for it should be transparently described. Because of international use of the term adaptation to refer to all processes summarized under adolopment (see GRADE-ADOLOPMENT below), the terms adolopment and adaptation are used as methods interchangeably when referring to guideline contextualization that reflects the process and concept of making recommendations relevant and implementable to the target setting. Adaptation of a recommendation is used to specifically refer to a single recommendation as opposed to adaptation of guidelines.

De novo development

De novo development involves formulating trustworthy recommendations for new prioritized questions that source guideline(s) do not answer. This can happen to fill gaps with recommendations that are particularly important for the target setting. For example, a subgroup of a population that is particularly important for the target setting but that has not been considered in source guidelines. The answers can be based on existing or new relevant evidence synthesis products (eg, systematic reviews or health technology assessments). De novo recommendations should include transparent descriptions of the certainty in the evidence and the judgements made on the EtD criteria to move from evidence to recommendations and follow best practice for guideline development.

Saudi Arabia (one), Tunisia (one), Asia-Pacific region (one), China (one), and Japan (one) (see Appendix 1) [27].

Given the experience with the approach and a recent initiative to develop a handbook for the WHO by the WHO Regional Office for Europe for guideline contextualization to strengthen countries, we further operationalized and refined the GRADE-ADOLOPMENT approach [7,14].

This update became necessary for the following reasons: a) the prior guidance was based on a single, albeit large, guideline development effort in one country, while the new guidance builds on case examples collected from around the globe; b) the experience with GRADE's EtDs frameworks , perhaps the most important contribution of the GRADE working group and central to GRADE ADOLOPMENT , has increased and led to more specific considerations about how they are included in this approach; c) the prior guidance did not undergo the GRADE working group's new and rigorous approval process, because this approach was developed after the publication of the previous article.

2. Objectives

This GRADE article provides updated guidance for using the GRADE-ADOLOPMENT approach to adopt and adapt recommendations from source guidelines and de novo develop recommendations when there are no existing recommendations.

3. Methods

3.1. Overview

We developed this GRADE guidance based on the original GRADE-ADOLOPMENT model and approach [7]. Also, it was informed by the WHO handbook for guideline adolopment [14], the review of guideline producing organizations' handbooks (R. N. M. El-Khoury et al, unpublished data, 2023), and results of a scoping review assessing guidelines that have used this approach [27]. Furthermore, authors of this article provided examples of GRADE-

ADOLOPMENT efforts to inform the operationalization. We integrated lessons learnt within the description of our stepwise approach.

3.2. Operationalization of guidance

We followed recently described methods for the development of GRADE guidance [28]. In brief, we used iterative discussions, online meetings, and email communications to produce draft-updated guidance. We then presented our approach and discussed it during the GRADE-ADOLOPMENT project group meetings held in July 2022 (20 people), September 2022 (15 people), October 2022 (17 people), November 2022 (20 people), March 2023 (23 people), and April 2023 (21 people). We presented the initial results to GRADE working group members for discussion and feedback in November 2022 (115 people) and in May 2023 (approximately 100 people) and twice to the GRADE working group in September 2023 (approximately 60 and 90 people). It was first approved at the Glasgow in the September 2023 meeting by approximately 89 members of the GRADE working group then by the GRADE guidance group on November

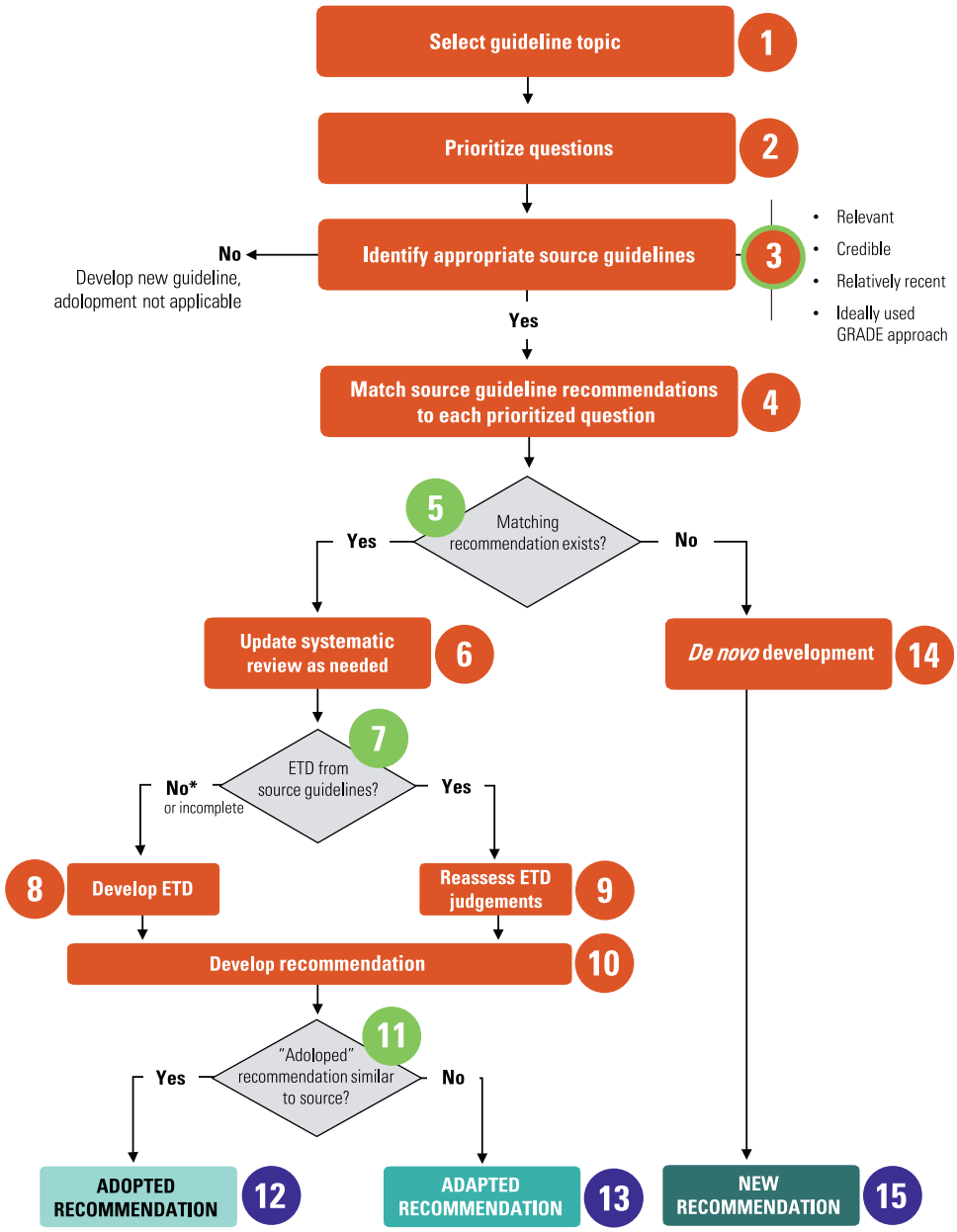

Figure 1. The Grading of Recommendations, Assessment, Development and Evaluations (GRADE)-ADOLOPMENT approach.Red circles mark general steps (1, 2, 4, 6, 8, 9, 10, 14), while the green circles indicate steps with decision points (3, 5, 7, 11), and blue circles label steps with specific outcomes of adolopment (12, 13, 15). * No EtD available (note that information that contributes to the decision criteria, eg, magnitude of benefits or estimate of values, may be described in the source guideline). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

22, 2023. Key discussion points revolved around clearly labeling decision nodes, clarifying the approach to using existing EtDs, providing more examples on the difference between adoption and adaptation, providing two new pathways for capturing relevant source guideline and matching recommendations, and updating the figure.

4. Results

Figure 1 provides an overview of the GRADE-ADOLOPMENT approach to contextualize guidelines. It describes the steps, decision points, and outcomes of adolopment. While these steps may occur as described, fewer steps may be taken by different groups, and some of the steps may occur in parallel or in another sequence, if appropriate (see Fig. 2). We provide a detailed description and examples on how each step can be operationalized by the guideline adoloper. The use of EtD criteria underpin this contextualization approach. Appendices 2 and 3 provide additional examples for each step and Appendix 4 a worked example for a single recommendation on breast cancer screening in the Czech Republic, based on European Commission guidelines [29,30].

The process described in Figure 1, and in detail below, leads to an adapted, adopted, or de novo (adoloped) recommendation. The adoloped recommendations on their own may be used for future adolopment efforts by others. When reporting on the process, authors should provide a summary of the changes between the original recommendation(s) and the modified ones, and a summary of the locally relevant evidence or updated EtD or evidence synthesis regardless of the outcome of adolopment, which can facilitate sharing and collaboration.

4.1. Step 1: select guideline topic

A group, institution, or organization prioritizes a topic or a general area of interest for the guideline, for example, screening for and diagnosis of tuberculosis. This can happen in response to a variety of impetuses, such as a stakeholder's needs assessment, a decision priority by a policy maker, a priority setting exercise, or a request for collaboration. This step often requires the establishment of a steering committee to oversee the process [31,32], and a clear list of prioritization criteria may be necessary to inform the process [32]. Thus, step 1 includes selecting a group that will be responsible for the development of the adoloped recommendations using established processes [33]. For example, for the Czech national guideline for breast cancer screening, an oversight committee of the national guideline project selected cancer screening as a priority topic and approved a multidisciplinary guideline panel [34].

After scoping and selection of a priority guideline topic, there are at least three entry points to GRADE-ADOLOPMENT that use different approaches to prioritizing questions (Fig. 2): selection of an existing guideline, question prioritization, or identification of a repository of recommendations, such as a recommendations map [35,36].

4.2. Step 2: prioritize questions

Guideline questions are based on contextual needs and priorities. A formal process to prioritize key health questions can involve reviewing what questions are addressed in existing guidelines, evidence syntheses, surveys of stakeholders, or prospective guideline panel members. For example, in the adolopment of guidelines in Saudi Arabia, guideline panel members completed online surveys to rate the relative importance of health questions for their country

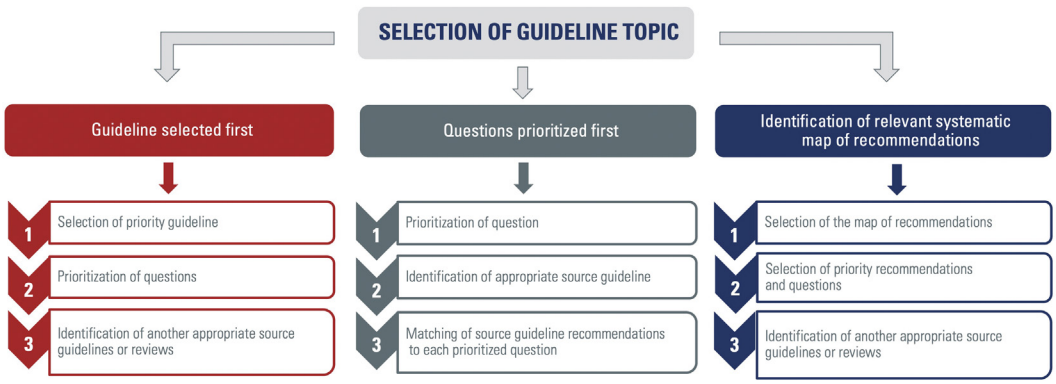

Figure 2. Alternative pathways for step 1 in the adolopment process.Developers can choose the pathway (vertically) that works best for them once they have selected the guideline topic. In the red pathway, they first select an existing guideline to adolop and then prioritize questions within this guideline. In the gray pathway, they prioritize questions first and then identify appropriate source guideline(s). In the blue pathway, they identify a systematic map of recommendations from which they will select the priority questions and recommendations to adolop. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

health care setting using a 9-point Likert-type scale (1 least important; 9 most important), from a source guideline (Fig. 2). This process was vetted by key stakeholders, such as health care actors from the Ministry of Health [7]. To ensure that the guidelines comprehensively addressed the selected topic with a complete set of recommendations, questions deemed complementary to those rated as important were also included. The preliminary questions were sent to panelists for approval, with opportunity for further input before finalization.

Similar approaches were used in other guideline adolopment processes in South America [11,37]. The prioritization process for questions does not differ importantly between an original guideline development effort and guideline adaptation projects. However, what distinguishes GRADE-ADOLOPMENT from simple guideline adaptation is the fact that the former focuses on priorities for the context as opposed to being constrained by what is present in one or multiple source guideline(s), which limits a guideline group to selecting questions from the source guidelines [32].

Thus, in GRADE-ADOLOPMENT, the selection of questions for single recommendations is driven by priorities of relevant stakeholders (ie, here are the questions, let us answer them through multiple existing guidelines or evidence syntheses, where possible). This priorities-driven approach allows guideline panels to include priority questions not addressed in previous guidelines and to broaden the scope (ie, the list of population, interventions, comparisons, and outcomes (PICO) questions), as described in the next steps [11,32]. In living guidelines where recommendations are added over time, guideline groups often need to consider both the feasibility of the work and the importance of the question when prioritizing recommendations [38−40]. The prioritization process also includes situations, where a single guideline is the only source, for example, a WHO or European Commission guideline. In this situation, panel members review the questions from the source guideline and decide whether to address all guideline questions, prioritize a subset, or add a question if missing. For example, in the Tunisian adolopment of the European Breast Guidelines (European Commission Initiative on Breast Cancer), the panel prioritized some of the questions and did not develop any recommendations de novo [29,30,41].

4.3. Step 3 and decision point: identify appropriate source guidelines

After identifying the scope (step 1) and prioritizing questions (step 2), a search for and assessment of existing guidelines or source of recommendations is required. What is searched will depend on the guideline group: the group may look in a single-source guideline, in multiple guidelines, or in a registry of recommendations, such as a recommendations map [35,36]. In the latter, the initial search should be broad and conducted in multiple databases and via other sources, such as websites or by contacting guideline developers [42]. Next, the identified guideline(s) or the guideline(s) publishing the identified recommendations may require evaluation to assess relevance (ie, address the topic of interest or some priority questions), credibility (for example, achieve more than 60% on the AGREE II instrument's domains of rigor of development and editorial independence [43−45]), recency (ie, an update would not be too much effort or the guideline is up to date), and assess, ideally, whether they are based on GRADE (ie, include evidence profiles or summary of findings (SoFs) tables and EtDs), which can facilitate the process. A review of 27 handbooks showed that guideline producers commonly use relevance, credibility, recency, and up-to-dateness as key criteria to evaluate existing guidelines. Recency relates to the concept of what new information, for example, from studies, is available since the source guideline recommendations and their supporting reviews were created. The more information is generated, the shorter is the acceptable time.

Additional criteria can also be considered at this stage when assessing the source guideline. Research evidence supporting the relative effects of interventions related to a particular recommendation is often *globally applicable*, whereas cost, values, equity, acceptability, and feasibility considerations are *usually not* globally applicable. Moreover, baseline risks for diseases can vary by context. Guideline developers can integrate the context specific baseline risks in the evidence summary tables when adoloping recommendations [41]. This can affect the absolute effect of an intervention and the evidence presented for benefits and harms and can lead to a different judgment by the panel. Thus, these factors in the source guideline should be assessed for directness (eg, similarity in baseline risk) prior to adopting or adapting recommendations. Reporting about which EtD criteria are context-specific will enhance transparency and allow use by others. Systematic reviews done to support a recommendation in a source guideline should be assessed for credibility, for example, with the risk of bias in systematic reviews (ROBIS) checklist or the Measurement Tool to Assess systematic Reviews (AMSTAR) 2 tool [46,47]. If the source for the guideline(s) already includes information about the recency and credibility, such as in a recommendation map, one can decide to use this information without repeating this assessment. If the source guideline is not based on systematic reviews, or when there are no independent or credible systematic reviews available, a formal recommendation can usually neither be adopted nor adapted in a trustworthy way. In such cases a guideline is typically deemed not credible for contextualization, and in this case, a new guideline needs to be developed. If however, parts of the

recommendation are supported by systematic reviews and the recommendation is deemed appropriate, additional evidence synthesis may be required for adaptation of a recommendation. Ideally, to facilitate the process of adolopment, the selected source recommendations were developed using GRADE and provided free access to evidence profiles or SoFs and EtDs in a compatible electronic format. If there are no existing recommendations then a de novo recommendation is required (step 15).

For example, when adoloping the Czech breast cancer screening guidelines, information specialists, and methodologists identified several existing breast cancer screening guidelines [34]. The guideline panel selected the European Breast Guidelines (on screening and diagnosis of breast cancer) as the source guideline for adolopment based on the scope, credibility, currency, feasibility, and applicability [29,30]. They adoloped all of the recommendations in the source guideline that are relevant for screening.

4.4. Step 4: match source guidelines recommendations to each prioritized question

In general, adolopment is driven by guideline questions that specify the PICO. For each prioritized guideline question, matching recommendations from the source guidelines(s) or recommendation map are identified in step 4 using the PICO components to understand the degree of concordance [48]. It is possible that more than one recommendation may be used to match a single prioritized question. For example, one may rely on multiple sources when not all prioritized outcomes are considered in a single source recommendation. Alternatively, it is possible that a new recommendation is required because none of the identified source guidelines includes a matching recommendation. For example, it is possible that some recommendations are from a WHO guideline, and others are from a National Institute for Health and Care Excellence (NICE) guideline. Alternatively, it is possible that a matching recommendation is not found in the original search, and a new search could be conducted. Capitalizing on existing collaborations or building collaborations early on in the process is valuable to facilitate the identification of matching recommendations that may not have been published yet or to identify other relevant evidence for adolopment [49,50]. For example, the American Gastroenterology Association (AGA) and Aga Khan University (AKU) shared information before the source guidelines were fully published to support the adolopment in Pakistan [51]. To accomplish this, requests for source material (eg, evidence profiles and EtDs) and sharing is required and that may involve using copyrighted material. Involving the source organization in the adolopment process to varying degrees further facilitates the process [41,52]. The adolopment group should check which material can be utilized (eg, depending on the type of publication) and may seek permission from the source guideline team or organization. Moreover, this step calls for meaningful engagement of those with an interest in the matter who would provide input on the context relevance of existing recommendations.

4.5. Step 5 and decision point: check whether a matching recommendation exists

If a matching source recommendation exists, one should determine whether one requires new or updated systematic reviews to contextualize them (see step 6). If no matching recommendation exists or there is no credible guideline that includes a corresponding recommendation, the corresponding prioritized question will require a de novo recommendation. For example, the Czech national guideline for breast cancer screening included one new recommendation because no source recommendation was appropriate [34]. Naturally, existing evidence syntheses can be used if available, for example, an existing Cochrane review (see step 3 and 15).

4.6. Step 6: update evidence synthesis as needed

This step reflects on whether existing evidence synthesis, that is, systematic reviews and evidence profiles, require updating or whether there is a need to conduct de novo evidence syntheses. The step requires an assessment of the underlying evidence for each recommendation across the benefits, harms, values, equity, resource, acceptability, and feasibility criteria, if all of these criteria are deemed necessary for the context. A representative panel of stakeholders could guide the judgment on which of these (if not all) are necessary. Ideally, all underlying evidence informing these EtD criteria should be based on a credible and up-to-date systematic review [53]. High quality, recent systematic reviews may not require updating or new synthesis that requires a judgment by people with methodological expertize. If reviews are outdated but useable, the literature searches should be updated to assess if new studies have been published and if they need to be integrated in existing evidence syntheses. There are other reasons for an update. For example, if the underlying evidence syntheses does not include all outcomes of interest for a prioritized question. It is also possible, that a source guideline recommendation is lacking information for an important EtD criterion. This will require obtaining that evidence, even if in the form of expert evidence [54]. The evidence summaries in the source guideline may also be too indirect in particular for the EtD criteria that consider baseline risk, values, or resources, which may require either additional or different evidence. For instance, a search for evidence about how people with the condition of interest value the outcomes in the target setting or a search for local cost information may be required [55].

The Czech breast cancer screening guideline group updated a systematic review and identified local evidence

for a recommendation [34] about the screening interval for asymptomatic women aged between 70 and 74 years. Czech data showed a 10% higher detection of breast cancer in 2-year intervals compared to 3-year intervals. Thus, the panel judged the balance of benefits and harms to be in favor of the shorter interval. The certainty of the evidence was unchanged (very low), but the recommendation changed direction from the source guideline.

4.7. Step 7 and decision point: check if EtD from source guideline is available

This step relates to checking if a related EtD exists in the source guideline. If the source guideline includes an EtD framework, the guideline panel should review the information and judgments in the source guideline and assess if evidence from the local context relevant needs to be integrated in the EtD table. Checking which EtD criteria are more important for the contextual setting (eg, resource use or feasibility) should happen in the beginning, but it could also take place at this step (eg, it may not be necessary to obtain information cost-effectiveness if this criterion is not relevant for the local setting). Moreover, the panel should explicitly report on which of these criteria predominantly guide their decision-making.

4.8. Step 8: develop EtD

Ideally, the adoloped guideline should include a completed EtD for each recommendation. If there is no EtD in the source guideline, then a new one should be developed even if it is based on information that is not systematically reported in a source guideline (step 8). This was the case for the adolopment in Saudi Arabia of the 2021 American College of Rheumatology source guidelines for the treatment of rheumatoid arthritis. A specific EtD framework was not available [56]. Information can often be hidden and not explicitly labeled as pertaining to a specific criterion, but if available, it can be used to create a new EtD. For example, the source guideline may include sections referring to equity or acceptability of an intervention as a narrative but not formally structured in an EtD framework.

Note that contextualization should be based on local evidence and not on the local processes in place (ie, existing circumstances do not determine the direction and strength of a recommendation − that should be based on evidence and may lead to recommendations that are contrary to local practice). Factors determining the use of specific management options, such as barriers and facilitators, may be addressed as implementation considerations. Sometimes, changes to legislation are necessary. For example, in an Estonian guideline for the use of HIV testing by trained lay people, existing legislation required that testing could be done by non-health professionals, which the law at that time did not permit [57]. The contextualization process provides opportunities to flag these feasibility issues and implementation considerations.

4.9. Step 9: reassess EtD judgements

If EtDs exist for the prioritized questions, then they need to be checked for completeness on all relevant EtD criteria, and the judgements in the source recommendation should be reviewed. When reassessing, agreement, or disagreement with the judgements should be noted for transparency and to allow any future adolopment. Reasons for disagreement should be described (eg, if there is new evidence that supports modifying a judgment, context-related variations in resources, local considerations of patient values, etc.). The adoloper should integrate the context-specific evidence under research evidence in the corresponding EtD criterion and make judgements based on this information accordingly.

Reassessment can take place in one of three ways:

1. The original guideline panel judgements are available, and the local panel decides whether they agree or not [41]. This approach has the advantage of saving time. However, it may limit the discussion, especially if panelists are new to the process. This option may be the most efficient alternative if the changes to the evidence and other considerations in the EtD framework are minimal.
2. The original guideline panel judgements are hidden, and the adolopment panel makes the judgements independently, using the available evidence [56]. The PanelVoice app is one way to collect input from the panel members before showing them the original judgements [58]. Additional details on PanelVoice can be found on https://gradepro.org/panel-voice/. This approach usually takes more time since panelists are required to discuss and reach agreement. It is probably the option that is least prone to influence by the source guideline.
3. A mixed approach combining the two previous alternatives. For example, some judgements may be left in place, while others may be open for discussion. If significant changes occur in the EtD framework due to contextual or new evidence or for domains that are a priori deemed as highly context dependent, then judgments in a source guideline might not be helpful and could potentially be misleading.

Whichever approach the adolopers use for one or for all the recommendations, they should report the approach explicitly in the adoloped guideline. For example, the Czech breast cancer guidelines used the first approach: the panel members had access to complete data and judgments from the European Commission in GRADEpro ahead of panel meetings. The group made changes to the judgements in only one recommendation, and agreed with the judgements for the other recommendations [34].

4.10. Step 10: develop recommendations

Steps 8 and 9 lead to complete and contextualized EtD frameworks. Of note, the process follows general best practices for guideline development. The panel should develop the recommendation and also complete the conclusion sections in the EtD as well as tailor them to the target context as needed. The direction and strength can be based on GRADE, regardless of the grading methodology of the original recommendation or guideline.

4.11. Step 11: "adoloped" recommendation similar to source

In this step, the adoloper checks if the recommendation is similar to the source recommendation. That is, if the resulting recommendation has the same population, interventions, and comparisons as the source recommendation, no change in the direction or strength of the recommendation, and no substantial changes to the information in the EtD, for example, in the evidence or judgments, it will be labeled as adopted (step 12).

For example, in the Latin American venous thromboembolism guidelines, the question of whether to offer prophylaxis with anticoagulants in people undergoing general surgery was considered. They maintained the direction and strength of the original recommendation (conditional in favor of prophylaxis), and made only some minor modifications to the remarks [41,59].

If the guideline development group decides that there are important changes to the recommendation, such as if a change is made in a judgment, if missing information in the EtD framework is inserted, or if the recommendation is different (eg, the population is narrower or broader), the recommendation will be labeled as adapted (step 13).

4.12. Step 12 and outcome: adopted recommendation

An adopted recommendation will contain no important changes to the recommendation itself or the information in the EtD. Changing the name of the development group, the order of the words, or using weak instead of conditional to label a recommendation is not an important change but should be done. For example, in the Tunisian breast cancer guidelines, five out of six recommendations were unmodified after the adolopment process [41].

The assessment and careful review of the existing recommendations and supporting evidence (including EtDs) is essential for adoption using the adolopment approach.

4.13. Step 13 and outcome: adapted recommendation

An adaptation is made when an adoloping group makes important changes to the recommendation, such as a modification to the text that changes the population, intervention, comparison, direction, or strength from the source guideline recommendation based on new evidence, judgements, or if there are important changes to the rationale, that is, the EtD supporting the recommendation. If the recommendation remains similar by chance, however, information in the EtD is new or judgments are different, an adapted recommendation is the result. Of note, if an adoloper makes strong recommendations, they should follow the GRADE guidance for generating strong recommendations and justify it appropriately [5,60].

For example, the 2015 American College of Rheumatology guidelines included a strong recommendation based on low-certainty evidence in favor of using a treat-to-target strategy rather than a nontargeted approach for early rheumatoid arthritis patients [61]. However, the adoloper made a conditional recommendation in the same direction with low-certainty evidence. The factors that influenced a conditional recommendation included different judgements around the balance of benefits and harms, moderate cost, and uncertainty about insurance coverage along with concerns for inequities, monitoring, and follow-up of patients [7,61,62]. The Latin American venous thromboembolism guideline group evaluated the use unfractionated heparin (UFH) or low-molecular-weight heparin (LMWH) in acutely critically and medically ill patients who require pharmacologic prophylaxis. The source guideline panel made a conditional recommendation in favor of LMWH, while the Latin American panel made a conditional recommendation in favor of either LMWH or UFH because of contextual affordability and availability [11]. The Reporting Tool for Adapted Guidelines in Health Care checklist has been developed specifically to improve the reporting of adaptation of guidelines and serves as a useful guide to report on the adolopment processs [63].

4.14. Step 14: de Novo development

If step 5 determined that there is no source recommendation or no credible guideline provides a recommendation, a new recommendation is required. The de novo development should follow trustworthy guideline development processes to arrive at a new recommendation, ideally including a new EtD. This development should be informed by existing evidence syntheses from systematic reviews or guidelines, but could require doing a new systematic review. If the adoloper uses existing systematic reviews, they should describe the EtD process by integrating local evidence, for example, related to cost and cost-effectiveness, and values, along with other evidence. If more than one credible systematic review is identified for a particular criterion on the EtD (eg, several reviews reporting on benefits and harms of an intervention), guideline groups may decide to

combine the data from the reviews. They could also conduct their own meta-analysis.

4.15. Step 15 and outcome: new recommendation

The adoloper should label a recommendation clearly as new, if applicable. For example, the Czech breast cancer screening guidelines included one new recommendation that was clearly labeled following a proper guideline development process documented using the original RIGHT reporting tool [64]. The recommendation was clearly marked as developed by the Czech guideline panel and not adopted or adapted from the source guideline.

4.16. Dissemination and implementation strategy

Contextualization efforts provide an opportunity to involve guideline development groups in the dissemination and implementation of guidelines. Examples of these interventions include leadership support and engagement, regular training and education, regular audit and feedback, engaging clinical and quality champions, and engaging patients as champions for change directly with adoloping guideline development group members. Guideline implementation tools should be tailored to the local or national context, including audit criteria and standards, clinical decision aid and mobile apps, and patient educational resources, among others [65,66]. The GIN implementation working group provides information on such tools, such as the clinician guideline determinants questionnaire and the guideline implementation planning checklist [67−69].

5. Discussion

In this GRADE guidance, we have further conceptualized and operationalized GRADE-ADOLOPMENT for adoption, adaptation, and de novo creation of guideline recommendations. It differentiates itself from other adaptation methods in a number of ways. First, it builds on the backbone of the actual criteria considered in an EtD framework. Second, it focuses on priorities for individual recommendations that are relevant for context as opposed to on a predetermined set of recommendations in a source guideline. Third, it allows for sequential use of evidence informing judgments on an EtD by multiple groups. This approach has been used in numerous guidelines, [61] and a number of factors facilitate its use, such as existing EtDs and detailed information in the source guideline about the determinants of the direction and strength of a recommendation, as well as sharing of information by source guideline developers. Sharing of information at the EtD level publicly allows for efficient contextualization using adolopment. Other factors create barriers, such as licensing cost related to source guidelines. Additional barriers to adolopment of guidelines are common to any adaptation approach, such as errors in the source guidelines, need for translation, lack of detailed information, and other factors. The article provides a number of improvements over our prior framework, including how source guideline recommendations can be selected, examples from a variety of adolopment projects that will help readers apply the approach, considerations about how to use existing and developing new EtDs, and clearer guidance on how to reassess the evidence and judgments in the EtD.

5.1. Strengths and limitations

This guidance builds on extensive field testing of the original approach and its global use by many independent groups. We identified at least 19 published adolopment projects (see appendix 1). Strengths of the approach relate to efficiency, contextual priorities, transparency, and the widely used EtD frameworks. The active involvement of guideline development group members in the evaluation of the source guideline leads to an increased sense of ownership. Limitations include the need for work that any guideline development group faces, such as having capacity and expertize in guideline methodology. We are not aware of high-quality existing evidence on whether guideline adaptation is more or less efficient compared to de novo development.

5.2. Implications for practice and research

Guideline groups and other decision makers can make their guideline development efforts more effective by using GRADE-ADOLOPMENT. To ensure quality, they should consider the following:

1. Check the 15 steps of GRADE-ADOLOPMENT for completion.
2. Report the guidelines using the Reporting Tool for Adapted Guidelines in Health Care reporting checklist [63].
3. Evaluate their group process and use of evidence in the adolopment, for example, by using the PANEL-VIEW instrument [70].
4. Include an independent evaluation of the project, for example, by using AGREE II.
5. External peer review.

Future research should explore if the adolopment process can be used in other efforts, such as health technology assessment reports and where opportunities lie to increase efficiency of contextualization efforts. Additional user testing and qualitative research to understand barriers and facilitators may help with further refinement.

6. Conclusions

In this GRADE guidance article, we describe the refined operationalization of GRADE-ADOLOPMENT based on 6 years of global experience and structured input from the GRADE working group, as well as published reports of external users.

CRediT authorship contribution statement

Miloslav Klugar: Writing − original draft, Methodology, Conceptualization. **Tamara Lotfi:** Writing − review & editing, Methodology. **Andrea J. Darzi:** Writing − review & editing, Methodology, Conceptualization. **Marge Reinap:** Writing − original draft, Methodology, Conceptualization. **Jitka Klugarová:** Writing − review & editing, Resources. **Lucia Kantorová:** Writing − review & editing, Resources. **Jun Xia:** Writing − review & editing. **Romina Brignardello-Petersen:** Writing − review & editing. **Andrea Pokorná:** Writing − review & editing, Resources. **Glen Hazlewood:** Writing − review & editing. **Zachary Munn:** Writing − review & editing. **Rebecca L. Morgan:** Writing − review & editing. **Ingrid Toews:** Writing − review & editing. **Ignacio Neumann:** Writing − review & editing. **Patraporn Bhatarasakoon:** Writing − review & editing. **Airton Tetelbom Stein:** Writing − review & editing. **Michael McCaul:** Writing − review & editing. **Alexander G. Mathioudakis:** Writing − review & editing. **Kristen E. D'Anci:** Writing − review & editing. **Grigorios I. Leontiadis:** Writing − review & editing. **Celeste Naude:** Writing − review & editing. **Lenny T. Vasanthan:** Writing − review & editing. **Joanne Khabsa:** Writing − review & editing. **Malgorzata M. Bala:** Writing − review & editing. **Reem Mustafa:** Writing − review & editing. **Karen DiValerio Gibbs:** Writing − review & editing. **Robby Nieuwlaat:** Writing − review & editing. **Nancy Santesso:** Writing − review & editing. **Dawid Pieper:** Writing − review & editing. **Saphia Mokrane:** Writing − review & editing. **Israa Soghier:** Writing − review & editing. **Wanchai Lertwatthanawilat:** Writing − review & editing. **Wojtek Wiercioch:** Writing − review & editing. **Shahnaz Sultan:** Writing − review & editing. **Jana Rozmarinová:** Writing − review & editing. **Pavla Drapačová:** Writing − review & editing. **Yang Song:** Writing − review & editing. **Marwa Amer:** Writing − review & editing. **Yasser S. Amer:** Writing − review & editing, Conceptualization. **Shahab Sayfi:** Writing − review & editing. **Ilse M. Verstijnen:** Writing − review & editing. **Ein-Soon Shin:** Writing − review & editing. **Zuleika Saz-Parkinson:** Writing − review & editing. **Kevin Pottie:** Writing − review & editing. **Alessandra Ruspi:** Writing − review & editing. **Ana Marušić:** Writing − review & editing. **K.M. Saif-Ur-Rahman:** Writing − review & editing. **Maria X. Rojas:** Writing − review & editing. **Elie A. Akl:** Writing − review & editing, Methodology, Conceptualization. **Holger J. Schünemann:** Writing − original draft, Methodology, Conceptualization.

Data availability

No original data.

Declaration of competing interest

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Supplementary data

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